

# K8S-10: Metadata Telemetry Enrichment

---

**Series:** K8S — Kubernetes Monitoring | **Notebook:** 10 of 13 | **Created:** January 2026 | **Last Updated:** 10/02/2026

## Enriching All Telemetry with Kubernetes Metadata

---

Kubernetes metadata enrichment automatically adds labels and annotations from your Kubernetes resources to all telemetry signals. This is the **recommended approach** for adding context to your observability data because it enriches everything: metrics, logs, traces, events, and entities.

---

## Table of Contents

---

1. [Why Metadata Enrichment?](#)
  2. [Enrichment Methods Comparison](#)
  3. [DynaKube Configuration](#)
  4. [Configuring Settings-Based Enrichment](#)
  5. [Namespace Selectors](#)
  6. [Understanding Enrichment Files](#)
  7. [Querying Enriched Data](#)
  8. [Use Cases](#)
  9. [Settings API](#)
  10. [Troubleshooting](#)
- 

## OneAgent Attribute Enrichment (OneAgent 1.333+)

**Requires:** OneAgent version **1.333** or later

OneAgent can enrich **all telemetry signals** (metrics, spans, logs, events, entities) with custom metadata at the source — before data reaches the Dynatrace platform. This is more efficient than server-side tagging (auto-tags) because enrichment happens on the host and propagates to all Smartscape nodes.

**Primary Fields** (standardized from Semantic Dictionary):

- `dt.security_context` — data governance and access control
- `dt.cost.costcenter` — cost allocation
- `dt.cost.product` — product attribution

**Primary Tags** (custom key-value pairs):

- `primary_tags.environment` — environment identification (production, staging, etc.)

- `primary_tags.team` — team ownership
- `primary_tags.business_unit` — organizational unit

**Configuration:**

```
# Set during OneAgent installation
Dynatrace-OneAgent-Linux.sh --set-host-tag="primary_tags.environment=production" --set-host-tag="dt.security_context=confidential"

# Set on existing agents via oneagentctl
oneagentctl --set-host-tag="primary_tags.environment=production"
oneagentctl --set-host-tag="dt.cost.costcenter=12345"

# Per-process via environment variable (overrides host-level)
DT_TAGS="primary_tags.team=platform primary_tags.environment=production"
```

**Benefits over auto-tagging:**

|                   |                                |                                                                       |
|-------------------|--------------------------------|-----------------------------------------------------------------------|
| Aspect            | Auto-Tags (Server-Side)        | Attribute Enrichment (Agent-Side)                                     |
| -----             | -----                          | -----                                                                 |
| When applied      | After data arrives at platform | At the source, before transmission                                    |
| Scope             | Entity tags only               | All signals: metrics, spans, logs, events, entities                   |
| Grail integration | Limited                        | Full — feeds OpenPipeline routing, bucket assignment, permissions     |
| Cost allocation   | Not supported                  | <code>dt.cost.costcenter</code> , <code>dt.cost.product</code> fields |
| Security context  | Not supported                  | <code>dt.security_context</code> for data governance                  |

See: [Primary Grail fields and tags enrichment through OneAgent](#)

## Prerequisites

| Requirement           | Details                                                  |
|-----------------------|----------------------------------------------------------|
| Dynatrace Environment | SaaS with Grail                                          |
| Permissions           | <code>settings.read</code> , <code>settings.write</code> |
| DynaKube              | Deployed with <code>metadataEnrichment</code> enabled    |
| Kubernetes            | Namespaces with labels/annotations to enrich             |

## 1. Why Metadata Enrichment?

Kubernetes metadata enrichment solves a critical observability challenge: **connecting telemetry to business context**.

### Common Use Cases

| Use Case        | Metadata Example                  | Benefit                  |
|-----------------|-----------------------------------|--------------------------|
| Cost allocation | <code>cost-center: finance</code> | Attribute costs to teams |

| Use Case                   | Metadata Example    | Benefit                      |
|----------------------------|---------------------|------------------------------|
| Environment identification | env: production     | Filter by deployment stage   |
| Team ownership             | team: checkout      | Route alerts to owners       |
| Compliance tagging         | compliance: pci-dss | Identify regulated workloads |
| Application grouping       | app: ecommerce      | Group related services       |

## What Gets Enriched

With settings-based enrichment, **all signals** receive metadata:

| Signal Type  | Enriched? | Notes                                 |
|--------------|-----------|---------------------------------------|
| Logs         | Yes       | All container and pod logs            |
| Metrics      | Yes       | Including Kubernetes platform metrics |
| Spans/Traces | Yes       | Distributed tracing data              |
| Events       | Yes       | Kubernetes events                     |
| Entities     | Yes       | Smartscape topology                   |

## Primary Grail Fields and Tags

In addition to settings-based enrichment, Dynatrace automatically propagates certain **Primary Grail Fields** across all signal types. These fields are indexed, require no configuration, and are the foundation for cross-signal data organization:

| Primary Grail Field  | DQL Field          | Propagation |
|----------------------|--------------------|-------------|
| Kubernetes Cluster   | k8s.cluster.name   | All signals |
| Kubernetes Namespace | k8s.namespace.name | All signals |
| Host Group           | dt.host_group.id   | All signals |
| AWS Account          | aws.account.id     | All signals |

**Primary Grail Tags** (prefix `primary_tags.`) extend this concept for custom dimensions like `primary_tags.app` or `primary_tags.team`. Use them when Primary Grail Fields don't align with your organizational structure (e.g., shared infrastructure hosting multiple applications).

### Setting Primary Tags Directly from Kubernetes (June 2026)

The [Kubernetes tag setup \(DT docs\)](#) page documents dedicated annotations for emitting primary tags at namespace or pod scope:

```
metadata:
  annotations:
    metadata.dynatrace.com/primary_tags.team: payments
    metadata.dynatrace.com/primary_tags.environment: production
```

Primary fields ride the same surface ( `metadata.dynatrace.com/dt.security_context` , `metadata.dynatrace.com/dt.cost.costcenter` , `metadata.dynatrace.com/dt.cost.product` ). Precedence is documented as pod over namespace: "When the same key is set at both scopes, the pod-level value wins for that pod." With Dynatrace Operator 1.10.0 the documented specificity chain extends to four levels (highest wins): pod annotations → namespace annotations → DynaKube resource attributes (see §3) → central configuration rules.

Two adoption caveats:

- **Version gates (corrected 07/31/2026).** These notebooks previously carried the quoted sentence "the at-source and central configuration options require minimum component versions: OneAgent version 1.343+, ActiveGate version 1.341+, Dynatrace Operator version 1.10+". That no longer matches the source pages, and it conflated two capabilities with different floors. The current published figures:

| Capability                                              | Minimum version                                    | Source                                                                                                                                                                            |
|---------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At-source / static tag enrichment (host tags, DT_TAGS ) | OneAgent 1.333+                                    | <a href="#">OneAgent tag setup (DT docs)</a> , <a href="#">OneAgent attribute enrichment (DT docs)</a>                                                                            |
| Central enrichment configuration                        | OneAgent 1.343                                     | <a href="#">What's new in OneAgent 1.343 (DT docs)</a> — "OneAgent now enriches telemetry data at the source based on a central enrichment configuration defined in the platform" |
| Kubernetes telemetry enrichment                         | Operator 1.10+, OneAgent 1.333+, ActiveGate 1.343+ | <a href="#">Kubernetes tag setup (DT docs)</a>                                                                                                                                    |

Two practical consequences. **At-source tagging is available far earlier than previously stated** — a 1.333–1.342 fleet can already emit primary tags at source and only lacks the *central* configuration surface. And the ActiveGate figure was **1.343+**, not 1.341+. OneAgent 1.343 has shipped. As always, tenant version is not agent version: verify your own fleet before relying on either floor.

- **Central configuration ships with SaaS 1.343 (July 2026 — staged tenant rollout):** rules that promote existing namespace labels to primary tags without manifest changes (limited to 20 rules per scope) arrive as *Centralized telemetry metadata enrichment* — key-value pairs, namespace annotations, and domain tags managed centrally. The docs previously flagged this as "the recommended approach" arriving mid-2026; it is the documented forward path alongside the settings-based enrichment described in this notebook — which remains the working path until 1.343 reaches your tenant (verify availability before designing against the central rules). On the ActiveGate side, AG 1.341 adds full namespace- and pod-level `metadata.dynatrace.com/primary_tags.<key>` enrichment and primary-Grail-field enrichment ( `dt.security_context` , `dt.cost.product` ) for Prometheus metrics monitored via ActiveGate. Note the remaining version gate — and note it supersedes the older figures quoted above: the at-source annotation path documents **OneAgent 1.333+**, **ActiveGate 1.343+**, and **Operator 1.10+** (shipped July 15, 2026), re-verified against the Kubernetes tag-setup page on 08/24/2026. The earlier "OneAgent 1.343+ / ActiveGate 1.341+" pairing was wrong in both directions: it over-stated the OneAgent floor and, more dangerously, under-stated the ActiveGate one.

**Cost allocation fields** like `dt.cost.costcenter` and `dt.cost.product` also propagate to service metrics, enabling chargeback reporting across teams.

**See also: ORGNZ-10: Advanced Segment Definitions** covers Primary Grail Fields in depth, including enrichment approaches for dedicated vs. shared infrastructure scenarios.

## 2. Enrichment Methods Comparison

Dynatrace offers multiple ways to add Kubernetes metadata. Choose based on your needs:

### Enrichment Methods Comparison

Settings-based (recommended) vs manual pod annotations

| Settings-Based (Recommended)                                                                                                                                                                                                                                                                                                                                                             | Manual Pod Annotations                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enriches ALL Signals</b><br><div>LogsMetricsTracesEvents</div> <div>Entities (Smartscape)</div> <ul style="list-style-type: none"><li>✓ Single configuration point</li><li>✓ Leverages namespace labels</li><li>✓ No application changes</li><li>✓ Works with K8s platform metrics</li><li>✓ Works with Prometheus metrics</li></ul> <div>Use this for comprehensive enrichment</div> | <b>Does NOT Enrich</b><br><div>Kubernetes MetricsKubernetes Events</div> <div>Smartscape EntitiesPrometheus Metrics</div> <ul style="list-style-type: none"><li>✗ Requires changes to every pod spec</li><li>✗ Only for application telemetry</li><li>✗ Manual annotation management</li><li>✗ Conflicts with settings-based mode</li><li>✗ 45-min propagation lag on changes</li></ul> <div>Use only as fallback when settings-based unavailable</div> |

**Recommendation**Configure rules in Settings > Cloud and virtualization > Kubernetes telemetry enrichment. Use namespace labels (team, env, cost-center). Do NOT mix settings-based and pod-annotation enrichment in the same cluster — see K8S-99 §7 Metadata Enrichment items 38–44.

### Why Settings-Based is Recommended

Settings-based enrichment:

- Enriches **all signals**: logs, metrics, traces, events, entities
- Single configuration point (Dynatrace Settings)
- Leverages existing namespace labels
- No application changes required
- Works with Kubernetes platform metrics

Manual pod annotations:

- Does NOT enrich: Kubernetes metrics, events, Smartscape entities, Prometheus metrics
- Requires changes to every pod spec

**Warning:** Do not mix settings-based enrichment with manual pod annotations. Using both simultaneously may cause conflicts and unexpected behavior.

**New (ActiveGate 1.345 — rollout from 08/25/2026): Kubernetes ingest enrichment supports custom rules.** The release note states plainly that *"ActiveGate now supports ingest enrichment for custom rules."* This adds a third lever alongside the DynaKube-level and settings-based methods compared above: enrichment applied at the ActiveGate on ingest, driven by your own rules rather than only the built-in Kubernetes metadata set. Where you already maintain enrichment in two places, check whether a custom ingest rule consolidates it — and mind precedence, since another enrichment source setting the same key still wins or loses by the ordering rules in §5. One documented limit from the same release note: *"Ingest enrichment rules with conditions are ignored by the ActiveGate."* — a conditional rule silently does nothing on this path, so keep conditional logic in the settings-based method.

The same release adds `container.runtime.name` to the Smartscape **CONTAINER** node — useful for fleets running more than one runtime (containerd, CRI-O, gVisor), because it makes "which runtime is this workload on" a queryable dimension rather than a node-labelling exercise. That matters in practice: gVisor is exactly the

runtime behind the `dynatrace-webhook` `CrashLoopBackOff` fixed in Operator 1.10.2, and this field is how you find those nodes before an upgrade.

Sources: [ActiveGate 1.345 release notes \(DT docs\)](#) — custom-rule ingest enrichment and the `container.runtime.name` field on the Smartscape `CONTAINER` node, read 08/28/2026.

### 3. DynaKube Configuration

Metadata enrichment must be enabled in your DynaKube custom resource.

#### Enable Metadata Enrichment

```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://ENVIRONMENT_ID.live.dynatrace.com/api

  # Enable metadata enrichment (enabled by default)
  metadataEnrichment:
    enabled: true

  oneAgent:
    cloudNativeFullStack: {}

  activeGate:
    capabilities:
      - kubernetes-monitoring
```

#### Disable Metadata Enrichment (if needed)

```
# Using annotation
kubectl annotate dynakube -n dynatrace dynakube \
  feature.dynatrace.com/disable-metadata-enrichment="true"
```

Or in the DynaKube spec:

```
spec:
  metadataEnrichment:
    enabled: false
```

#### Defining Resource Attributes in the DynaKube (Operator 1.10.0+)

Dynatrace Operator **1.10.0** (released July 15, 2026) adds a cluster-scoped enrichment surface: static resource attributes defined directly in the DynaKube spec. The [Kubernetes tag setup \(DT docs\)](#) page positions this mechanism in the primary-tag specificity chain below pod/namespace annotations and above central configuration rules. Three spec fields participate:

| Spec field                                                                | Applies to                                               | Precedence                                                            |
|---------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|
| <code>.spec.resourceAttributes</code>                                     | All signals (OneAgent, OTLP, log monitoring, ActiveGate) | Base — overridden by the mode-specific fields below on duplicate keys |
| <code>.spec.oneAgent.&lt;mode&gt;.additionalResourceAttributes</code>     | OneAgent-emitted signals only                            | Wins over <code>.spec.resourceAttributes</code> on duplicate keys     |
| <code>.spec.otlpExporterConfiguration.additionalResourceAttributes</code> | OTLP telemetry only                                      | Wins over <code>.spec.resourceAttributes</code> on duplicate keys     |

```
spec:
  resourceAttributes:
    aws.account.id: "123456789012"
  oneAgent:
    cloudNativeFullStack:
      additionalResourceAttributes:
        my.team: platform
  otlpExporterConfiguration:
    additionalResourceAttributes:
      my.team: platform
```

Constraints documented with the feature:

- A **soft limit of 10 attributes** applies across `.spec.resourceAttributes` and all `additionalResourceAttributes` blocks combined.
- Keys are sanitized (invalid DNS characters replaced); keys over 63 characters violate Kubernetes label limits.
- When both OneAgent and OTLP injection are active on the same pod, **conflicting keys produce undefined behavior** — keep the two mode-specific blocks consistent.
- The values are cluster-scoped and static. Use pod/namespace annotations for per-workload values; use this surface for cluster-wide facts (account IDs, environment, owning platform team).

**Running an earlier Operator?** These spec fields require Operator 1.10.0+ — on earlier versions the DynaKube rejects them at admission. The namespace/pod annotation path and the settings-based enrichment described in this notebook remain the working paths until your clusters upgrade.

## Tagging Rule Precedence Changed in Operator 1.10.2

**Operator 1.10.2** (released July 30, 2026) fixed a regression in Kubernetes workload and namespace tagging where, when several rules matched the same key, **each subsequent matching rule overwrote the previous one**. From 1.10.2, **only the first matching rule for a given key applies**.

This is a **behaviour change, not only a defect fix**, and it is the one thing to check before upgrading:

| Your rule set            | Effect of upgrading to 1.10.2      |
|--------------------------|------------------------------------|
| At most one rule per key | None. The fix is invisible to you. |

| Your rule set                                                                      | Effect of upgrading to 1.10.2                                                                             |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Several rules matching the same key, and you relied on the <b>last</b> one winning | <b>Tag values change.</b> The first matching rule now wins instead. Audit and reorder before you upgrade. |
| Several rules matching the same key, ordering incidental                           | Values may change; confirm the winner is the one you want.                                                |

Because these tags flow into `dt.security_context`, cost fields and primary tags, a silent flip changes what IAM boundaries admit and how cost is attributed — the failure is in *routing and access*, not in an error message. Audit first:

```
# List the enrichment/tagging rules currently configured
kubectl -n dynatrace get dynakube -o yaml | grep -A20 "resourceAttributes"

# After upgrading, confirm the resulting values on live telemetry
```

```
// Which security contexts and primary tags are actually landing on telemetry?
fetch logs, from:-1h
| filter isNotNull(dt.security_context)
| summarize recordCount = count(), by:{k8s.cluster.name, k8s.namespace.name, dt.security_context}
| sort recordCount desc
| limit 50
```

The same release also **starts logging metadata-enrichment rules the Operator cannot apply** instead of disregarding them silently — so after upgrading, the operator log is worth reading once for rules that were never taking effect:

```
kubectl -n dynatrace logs -l app.kubernetes.io/name=dynatrace-operator --tail=200 | grep -i enrich
```

**Adopting on your own schedule.** Operator upgrades roll out per estate, not per tenant. Until 1.10.2 reaches a given cluster, the previous last-rule-wins behaviour is what that cluster exhibits — and everything else in this notebook describes it correctly. Plan the rule audit as part of the upgrade, not after it.

**See:** [Metadata enrichment \(DT docs\)](#), [Operator 1.10.0 release notes \(DT docs\)](#), [Operator 1.10.2 release notes \(DT docs\)](#)

## 4. Configuring Settings-Based Enrichment

### Access Settings

1. Navigate to **Settings > Cloud and virtualization > Kubernetes telemetry enrichment**
2. Select **Add rule**

### Rule Configuration Options

| Field                | Description                       | Example            |
|----------------------|-----------------------------------|--------------------|
| <b>Metadata type</b> | Source type (annotation or label) | <code>Label</code> |

| Field  | Description                                | Example |
|--------|--------------------------------------------|---------|
| Key    | The label/annotation key to capture        | team    |
| Prefix | Optional prefix for the enriched attribute | k8s.    |

## Example: Capture Team Label

If your namespaces have:

```
apiVersion: v1
kind: Namespace
metadata:
  name: checkout
  labels:
    team: checkout-team
    cost-center: engineering
    env: production
```

Create enrichment rules:

| Rule | Metadata Type | Key         | Prefix |
|------|---------------|-------------|--------|
| 1    | Label         | team        | k8s.   |
| 2    | Label         | cost-center | k8s.   |
| 3    | Label         | env         | k8s.   |

Result: All telemetry from the checkout namespace will have:

- k8s.team: checkout-team
- k8s.cost-center: engineering
- k8s.env: production

## 5. Namespace Selectors

By default, enrichment rules apply to **all namespaces**. Use namespace selectors to limit scope.

### DynaKube Namespace Selector

If your DynaKube uses a namespaceSelector, ensure it matches the namespaces you want to enrich:

```
spec:
  metadataEnrichment:
    enabled: true
    namespaceSelector:
      matchLabels:
        dynatrace-enrich: enabled
```

## Labeling Namespaces for Enrichment

```
# Enable enrichment for a namespace
kubectl label namespace checkout dynatrace-enrich=enabled

# Verify labels
kubectl get namespace checkout --show-labels
```

## Match Expressions

For more complex selection:

```
namespaceSelector:
  matchExpressions:
    - key: env
      operator: In
      values:
        - production
        - staging
```

## 6. Understanding Enrichment Files

The Dynatrace Operator creates enrichment files that are used by OneAgent and applications.

### File Locations

| File                   | Location                       | Purpose              |
|------------------------|--------------------------------|----------------------|
| dt_metadata.json       | /var/lib/dynatrace/enrichment/ | JSON format metadata |
| dt_metadata.properties | /var/lib/dynatrace/enrichment/ | Properties format    |
| dt_host_metadata.json  | /var/lib/dynatrace/enrichment/ | Host-level metadata  |

### Example dt\_metadata.json Content

```
{
  "k8s.namespace.name": "checkout",
  "k8s.pod.name": "checkout-api-7d9f8c6b4d-x2k9m",
  "k8s.team": "checkout-team",
  "k8s.cost-center": "engineering",
  "k8s.env": "production"
}
```

## Accessing Enrichment Files in Applications

For manual instrumentation or custom applications:

```
import json
import os

# Load Dynatrace metadata
enrichment_paths = [
    '/var/lib/dynatrace/enrichment/dt_metadata.json',
    '/var/lib/dynatrace/enrichment/dt_host_metadata.json'
]

metadata = {}
for path in enrichment_paths:
    if os.path.exists(path):
        with open(path) as f:
            metadata.update(json.load(f))

print(metadata)
```

## 7. Querying Enriched Data

---

Once enrichment is configured, you can filter and group by the enriched attributes.

```
// Query logs by enriched team label
fetch logs, from:-1h
| filter isNotNull(k8s.namespace.name)
| summarize logCount = count(), by:{k8s.namespace.name}
| sort logCount desc
| limit 20
```

```
// Group metrics by cost center (enriched label) – sum() of every container in the namespace
// rollup: avg averages each container within a time bucket before the cross-container sum;
// without it, sum() also adds the 1-min points in each bucket and wide timeframes read ~10x high.
timeseries cpuMillicores = sum(dt.kubernetes.container.cpu_usage, rollup: avg), from:-1h, by:
{k8s.namespace.name}
| fieldsAdd avgCpuMillicores = arrayAvg(cpuMillicores)
| sort avgCpuMillicores desc
```

```
// Find all spans from production environment
fetch spans, from:-1h
| filter isNotNull(k8s.namespace.name)
| summarize
    spanCount = count(),
    avgDuration = avg(duration),
    by:{k8s.namespace.name, service.name}
| sort spanCount desc
| limit 20
```

## 8. Use Cases

---

### Cost Allocation by Team

Label namespaces with cost centers:

```
kubectl label namespace checkout cost-center=checkout-team
kubectl label namespace catalog cost-center=catalog-team
kubectl label namespace shared cost-center=platform
```

Create enrichment rule for `cost-center` label, then query:

```
timeseries totalCpu = sum(dt.kubernetes.container.cpu_usage, rollup: avg), from:-1h, by:{"k8s.cost-center"} // backticks required – the hyphen is not a valid bare identifier
```

## Pipeline Routing (Bucket Assignment)

Send logs to different buckets based on enriched metadata. OpenPipeline has no "route" processor for this: inside a pipeline, the bucket is chosen in the **Bucket assignment** stage, where each **Bucket assignment** processor pairs a DQL matching condition with a target bucket. Add one processor per destination:

| Matching condition (DQL)             | Bucket                        |
|--------------------------------------|-------------------------------|
| <code>k8s.env == "production"</code> | <code>prod_logs_365d</code>   |
| <code>k8s.env == "staging"</code>    | <code>staging_logs_35d</code> |

The stage applies the first matching processor only, so order processors from most to least specific. Matching conditions are DQL: strings take double quotes, and a single-quoted condition such as `k8s.env == 'production'` is rejected.

## Security Context Assignment

Use enriched metadata in security context for fine-grained access. This happens in the pipeline's **Permission** stage: add a **Set security context** processor with the matching condition `k8s.team == "checkout-team"` and the value `team:checkout`.

**Sources:** [Processing in OpenPipeline \(DT docs\)](#) — Bucket assignment stage: "Assign records to the best fit bucket." Permission stage: "Apply security context to the records that match the query."

## Grail Permissions

Create IAM policies based on Kubernetes attributes:

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:logs:read WHERE storage:k8s.namespace.name = "checkout";
```

## 9. Settings API

Manage enrichment rules programmatically via the Settings API.

### Schema

The schema for Kubernetes telemetry enrichment is: `builtin:kubernetes.generic.metadata.enrichment`

**This schema is being replaced by the central configuration (SaaS 1.345+ — verify your tenant).** Starting with SaaS 1.345 you can opt a cluster, or the whole tenant, into central configuration, and the rules move to `builtin:ingest.enrichment.config`. The examples below still describe the working path until you opt in.

What to know before you opt in:

- **Version floors.** The migration guide lists *"Dynatrace Operator version 1.10 or later", "OneAgent version 1.333 or later", "ActiveGate version 1.343 or later", and "Dynatrace platform version 1.345 or later (required for the central configuration opt-in)".* Primary Grail tags on edge additionally need *"Dynatrace platform version 1.348 or later and ActiveGate version 1.345 or later"*. SaaS 1.348 is a staged rollout, so check it has reached your tenant.
- **Automatic migration happens only once.** When you opt in, *"If none exist, your current enrichment setup is automatically migrated to the new central configuration setting in the background. If rules already exist, no automatic migration occurs and your existing setup remains unchanged."* If you create even one rule in the new schema by hand first, you migrate the rest yourself.
- **Inheritance flips from override to merge.** In the current schema, *"rules defined at the Kubernetes cluster level replace rules defined at the environment level."* In the new schema, *"The new settings merge rules from the environment level with rules from the cluster level, where cluster-level rules take precedence."* So an environment-level rule that a cluster-level rule set silently hid today starts applying to that cluster after migration. Review environment-level rules before you opt in, because tags and cost fields can appear where they never did.
- **Automation that writes this schema** (the `curl` examples below, Monaco, Terraform) must move to the new schema id. The new rules use a different shape as well: the guide maps a current `Label` rule to a `K8S_NAMESPACE_LABEL` rule with a `target`.

The ready-made *Check your upgrade readiness* dashboard lists clusters whose enrichment rules have not been migrated yet.

**Sources:** [Migrate to central configuration for Kubernetes telemetry enrichment \(DT docs\)](#) — all quotes in this callout.

## List Current Rules

```
curl -X GET "https://ENVIRONMENT_ID.live.dynatrace.com/api/v2/settings/objects" \
-H "Authorization: Api-Token YOUR_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "schemaIds": ["builtin:kubernetes.generic.metadata.enrichment"],
  "scopes": ["environment"]
}'
```

## Create Rule via API

```
curl -X POST "https://ENVIRONMENT_ID.live.dynatrace.com/api/v2/settings/objects" \
-H "Authorization: Api-Token YOUR_TOKEN" \
-H "Content-Type: application/json" \
-d '[
  {
    "schemaId": "builtin:kubernetes.generic.metadata.enrichment",
    "scope": "environment",
    "value": {
      "enabled": true,
      "metadataType": "LABEL",
      "key": "team",
      "prefix": "k8s."
    }
  }
]
```

## 10. Troubleshooting

### Enrichment Not Appearing

| Symptom                 | Cause                       | Solution                            |
|-------------------------|-----------------------------|-------------------------------------|
| No enriched attributes  | Rules not propagated        | Wait 45 minutes after rule creation |
| Some namespaces missing | namespaceSelector mismatch  | Verify DynaKube namespaceSelector   |
| Pods not enriched       | metadataEnrichment disabled | Check DynaKube spec                 |

### Verify DynaKube Status

```
# Check DynaKube configuration
kubectl -n dynatrace get dynakube -o yaml | grep -A5 metadataEnrichment

# Check for disable annotation
kubectl -n dynatrace get dynakube -o yaml | grep disable-metadata-enrichment
```

### Verify Enrichment Files

```
# Check enrichment directory exists
kubectl exec -it <pod-name> -- ls -la /var/lib/dynatrace/enrichment/

# View enrichment content
kubectl exec -it <pod-name> -- cat /var/lib/dynatrace/enrichment/dt_metadata.json
```

### Timing Considerations

| Action           | Propagation Time |
|------------------|------------------|
| New rule created | Up to 45 minutes |

| Action                  | Propagation Time           |
|-------------------------|----------------------------|
| Rule modified           | Up to 45 minutes           |
| DynaKube deployed first | Immediate (if rules exist) |

**Tip:** If you configure enrichment rules **before** deploying DynaKube, the rules take effect immediately when DynaKube is applied.

## Summary

In this notebook, you learned:

- Why settings-based metadata enrichment is the recommended approach
- How to enable enrichment in DynaKube
- Creating enrichment rules in Settings UI
- Using namespace selectors to scope enrichment
- Understanding enrichment file locations and formats
- Querying enriched data with DQL
- Practical use cases: cost allocation, routing, security
- Troubleshooting common issues

## References

- [Metadata enrichment for K8s telemetry \(DT docs\)](#)
- [Configure enrichment directory \(DT docs\)](#)
- [Settings API — K8s Telemetry Enrichment schema \(DT docs\)](#)
- [K8s security context Grail permissions \(DT docs\)](#)
- [Set up Dynatrace on Kubernetes \(DT docs\)](#)
- [Kubernetes tag setup \(DT docs\)](#)
- [DynaKube parameters \(DT docs\)](#)

*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*